

ODAN: An Object-driven AR Narrative Framework

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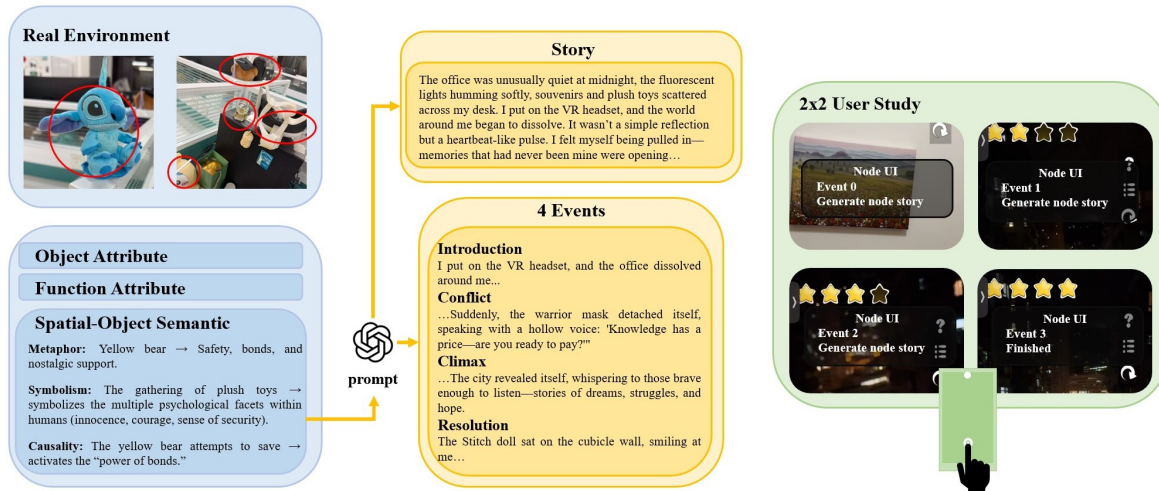


Figure 1: Using Spatial-Object Semantics, we decompose scene objects into layered meaning—base attributes and functions plus higher-level semantics. We inject these layers into generation prompts to produce a full storyline with 4 constituent events.

ABSTRACT

In adaptive AR storytelling, many systems still rely on coarse object labels and fixed spatial anchors, treating environments as static backdrops and overlooking semantic depth in how objects relate. We present an object-driven AR story framework that integrates Vision-Language Models (VLMs) with a mediating layer we call Spatial-Object Semantics, allowing nuanced narrative cues aligned with the scene in real time. Experimental results show that guiding VLMs with appropriate layered semantics transforms AR scenes from static settings into dynamic, story-rich spaces, coordinating perception, context, and plot while preserving real-time responsiveness, marking a step toward more expressive, responsive AR narratives.

Index Terms: Augmented Reality, Storytelling, Vision Language Model

1 INTRODUCTION

As a spatial narrative medium, Augmented Reality (AR) has become increasingly influential in education, cultural heritage, entertainment [4], where narrative design is central to engagement and user experience. AR evolved from enabling stable spatial overlays [1] to object- and site-anchored storytelling [3], and now to semantics-

and layout-aware narrative composition where story structure is co-designed with physical space [6]. Yet most AR narrative systems rely on surface semantics (for example, object tags or context triggers), which can reduce rich scenes to mechanical tasks. Associating a “chair” with “sit” is straightforward [7], but such mappings miss the symbolic and cultural resonance of spatial arrangements (e.g., pill bottles and a wedding ring on a nightstand). Consequently, current systems under-utilize semantic depth, limiting narrative coherence and emotional impact.

To address this challenge, we propose a system to automatically generate interactive AR stories with the help of Vision Language Models (VLMs) such as GPT-4o [8]. We decompose object meaning into object, function, and semantic attribute, which we called Spatial-Object Semantic, allowing more meaningful AR storytelling. This system includes an object-driven ar story generation pipeline, enabling users without any storytelling experience could create a unique ar story with current real environment. The user study indicates that narratives enhanced with Spatial-Object Semantics outperformed those without it across multiple dimensions, though not consistently for all measures.

2 SYSTEM DESIGN

Our system targets short, object-driven story generation situated in AR. We therefore adopt a prompt-based generation paradigm: prior work on LLM-based story generation and narrative evaluation suggests that short stories can be elicited effectively via zero-shot or light few-shot prompting [5].

Building on this premise, we introduce a Spatial-Object Semantics intermediary to better ground narratives in real-world scenes. Spatial-Object Semantics operationalizes scene grounding along three coupled dimensions that support narrative sense-making: (i) causal feasibility, (ii) spatial context, and (iii) symbolic/metaphoric significance. The causal dimension is motivated by classic findings

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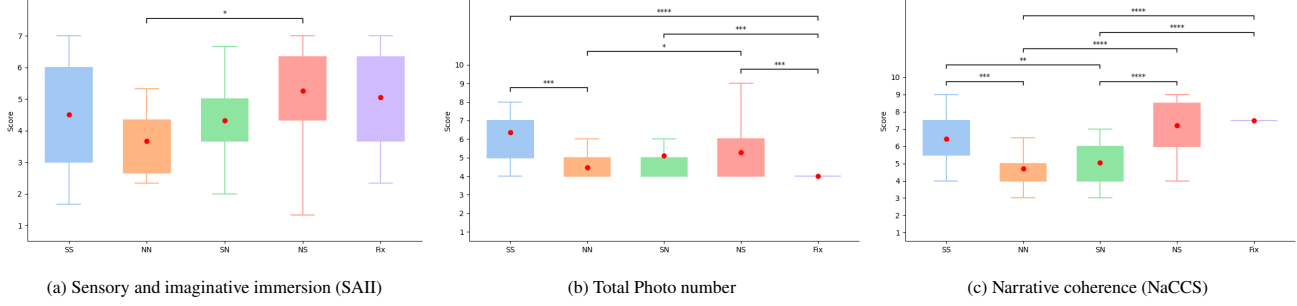


Figure 2: The statistical analysis of different evaluation results. (* $p < .05$, ** $p < .01$, *** $p < .001$)

that causal connectivity underpins narrative coherence and memory [9]. The spatial dimension draws on place-based accounts arguing that locations carry histories and atmospheres that contribute narrative meaning [6]. Finally, the symbolic/metaphoric dimension is grounded in semiotics and multimodal metaphor theory, where objects and configurations can cue themes and affect through denotation, connotation and metaphorical mapping [2].

3 STUDY DESIGN AND RESULTS

We conducted a 2×2 factorial user study that manipulated whether spatial-object semantics were used in (i) story content and (ii) story structure. This yielded four conditions: SS (content and structure), SN (content only), NS (structure only), and NN (label-only semantics). We also included a Fixed baseline with a predefined, non-object-driven story.

We tested one overarching hypothesis:

H. Narratives with a semantic intermediary layer (SS, SN, NS) will score higher than NN on understandability (H1), sensory and imaginative immersion (H2), exploration behavior (H3), narrative coherence (H4), and story-place semantic fitness (H5).

Narrative Quality was measured across four facets: narrative understanding (NES), sensory and imaginative immersion (GEQ), narrative coherence (NaCCS), and objective engagement (number of photos taken). Story-Place Semantic Fitness was rated with items adapted from prior AR narrative work [7, 6] to capture the ecological validity and semantic plausibility of placing story elements in the observed environment.

We recruited 17 volunteers (6 female, 11 male), aged 20–33. To ensure comprehension, all narratives were generated in each participant’s native language.

Overall, semantic intermediary conditions outperformed NN on several outcomes, but effects were not consistent across all measures. Understandability did not differ across conditions ($p = 0.440$). Immersion differed significantly ($p = 0.039$): NS scored higher than NN, providing partial support for H2.

Exploration behavior showed strong effects of object-driven narrative dynamics. All object-driven conditions produced more photos than the Fixed baseline ($p < 0.001$). In addition, SS and NS produced more photos than NN ($p = 0.032$, $p = 0.041$), suggesting that structural semantics particularly encourages environmental exploration (partial support for H3).

Narrative coherence differed significantly across conditions ($p < 0.001$). SS and NS scored higher than NN and SN, indicating that applying semantics to story structure is important for logical and consistent story progression (partial support for H4).

Story-place semantic fitness did not differ among SS, SN, NS, and NN. However, all object-driven conditions scored higher than the Fixed baseline ($p < 0.001$), suggesting that environmental responsiveness alone improves story-place congruence regardless of semantic depth.

4 CONCLUSION

We propose ODAN to abstract the rich semantic information with VLMs to advance AR storytelling beyond surface-level object tags and static spatial anchors. By utilizing the physical, functional, and symbolic roles of objects, this framework provides richer cues for narrative generation and alignment with the environment. Results from a user study demonstrated that semantic integration enhanced immersion, coherence, and exploratory behaviors, with structural-level semantics yielding the strongest effects. These findings suggest that AR can evolve from object label-driven content placement toward semantics-driven storytelling, positioning everyday spaces as dynamic, meaning-rich narrative media.

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